

Report Period	Sep 26, 2026 20:14 - Sep 26, 2026 20:20
Generated At	2026-09-26 20:20:15
Classification	ML: Isolation Forest + Random Forest

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16727	
Total Threats Mitigated	25	Total Packets Analyzed		5473818033	
High Priority	2	Malicious Packets		5457639871	
Low Priority	23	True Negatives Passed		10321	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		17	
False Positives	216	UDP Flood		2	
FP Rate	1.68%	Uncertain		6	
IF Precision	97.28%				
IF Recall	95.87%	Action		Count	
IF F1-Score	96.57%	Quarantined		4	
IF Accuracy	97.37%	Time Ban		4	
		Blackhole		0	
		Released		4	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	97.28%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.87%	Share of actual attacks successfully flagged
F1-Score	96.57%	Balanced measure combining Precision and Recall
Accuracy	97.37%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.68%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.13%	Actual attacks that went undetected
True Positive Rate (TPR)	95.87%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.32%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 6150	FN 265
Actual: Normal	FP 172	TN 10056

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.98%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.99%	Balanced measure combining Precision and Recall
Accuracy	99.98%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.58%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.42%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	6150	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	0

2c. Hybrid Cascaded

Metric	Value	Description
Precision	97.28%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.87%	Share of actual attacks successfully flagged
F1-Score	96.57%	Balanced measure combining Precision and Recall
Accuracy	97.37%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	1.68%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.13%	Actual attacks that went undetected
True Positive Rate (TPR)	95.87%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	98.32%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 6150	FN 265
Actual: Normal	FP 172	TN 10056

2d. Response Latency

Metric	Value	Description
Detection Time	524.7 ms	Average time taken to detect an attack
Mitigation Response Time	4.3 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.19%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	28.37%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	17.85%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.16	4	3	2026-09-26 20:17:08	2026-09-26 20:19:49
10.0.0.17	4	3	2026-09-26 20:17:13	2026-09-26 20:19:53
10.0.0.19	4	3	2026-09-26 20:17:10	2026-09-26 20:19:51
10.0.0.18	5	1	2026-09-26 20:17:10	2026-09-26 20:19:50
10.0.0.10	1	1	2026-09-26 20:17:31	2026-09-26 20:18:01
10.0.0.5	8	1	2026-09-26 20:17:26	2026-09-26 20:20:08
10.0.0.8	1	1	2026-09-26 20:16:18	2026-09-26 20:16:29

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 20:16:29	10.0.0.8	DDoS	Uncertain	68.7%	Low	Released
2026-09-26 20:17:08	10.0.0.16	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:17:10	10.0.0.18	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:17:10	10.0.0.19	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:17:13	10.0.0.17	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 20:17:18	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (30s)
2026-09-26 20:17:19	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban
2026-09-26 20:17:20	10.0.0.18	DDoS	SYN Flood	91.4%	Critical	Time Ban (30s)
2026-09-26 20:17:21	10.0.0.18	DDoS	SYN Flood	91.8%	Critical	Time Ban
2026-09-26 20:17:21	10.0.0.19	DDoS	SYN Flood	90.3%	Critical	Time Ban (30s)
2026-09-26 20:17:22	10.0.0.19	DDoS	SYN Flood	90.3%	Critical	Time Ban
2026-09-26 20:17:24	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban (30s)
2026-09-26 20:17:24	10.0.0.17	DDoS	SYN Flood	90.9%	Critical	Time Ban
2026-09-26 20:17:31	10.0.0.10	DDoS	UDP Flood	92.0%	High	Time Ban (30s)
2026-09-26 20:17:37	10.0.0.5	DDoS	Uncertain	58.9%	Low	Released
2026-09-26 20:17:51	10.0.0.18	DDoS	SYN Flood	94.0%	Critical	Released
2026-09-26 20:18:01	10.0.0.10	DDoS	UDP Flood	92.0%	High	Released
2026-09-26 20:18:01	10.0.0.18	DDoS	SYN Flood	94.0%	Critical	Time Ban (1m)
2026-09-26 20:18:19	10.0.0.16	DDoS	SYN Flood	92.9%	Critical	Time Ban (1m)
2026-09-26 20:18:22	10.0.0.19	DDoS	SYN Flood	94.1%	Critical	Time Ban (1m)
2026-09-26 20:18:24	10.0.0.17	DDoS	SYN Flood	93.2%	Critical	Time Ban (1m)
2026-09-26 20:19:49	10.0.0.16	DDoS	SYN Flood	93.4%	Critical	Time Ban (2m)
2026-09-26 20:19:50	10.0.0.18	DDoS	SYN Flood	93.1%	Critical	Time Ban (5m)
2026-09-26 20:19:51	10.0.0.19	DDoS	SYN Flood	91.9%	Critical	Time Ban (2m)
2026-09-26 20:19:53	10.0.0.17	DDoS	SYN Flood	91.4%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.5	Uncertain	0.6049	61.8%	Medium	Phase 1	1	11s	2026-09-26 20:20:08	Attack Stopped
10.0.0.17	SYN Flood	0.6279	91.4%	Critical	Phase 2	1	11s	2026-09-26 20:19:53	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6449	91.9%	Critical	Phase 2	1	11s	2026-09-26 20:19:51	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6445	93.1%	Critical	Phase 2	1	10s	2026-09-26 20:19:50	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6380	93.4%	Critical	Phase 2	1	11s	2026-09-26 20:19:49	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.5973	61.8%	Medium	Phase 1	1	11s	2026-09-26 20:19:43	Attack Stopped
10.0.0.5	Uncertain	0.6065	61.8%	Low	Phase 1	1	10s	2026-09-26 20:19:29	Attack Stopped
10.0.0.18	SYN Flood	0.6235	95.8%	Critical	Phase 2	1	1m 10s	2026-09-26 20:19:01	Ban Expired
10.0.0.17	SYN Flood	0.6389	93.2%	Critical	Phase 2	3	1m 11s	2026-09-26 20:18:24	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6376	61.9%	Medium	Phase 1	1	10s	2026-09-26 20:18:22	Attack Stopped
10.0.0.19	SYN Flood	0.6428	94.1%	Critical	Phase 2	3	1m 12s	2026-09-26 20:18:22	Attack persisted - escalating to time ban
10.0.0.16	SYN Flood	0.6275	92.9%	Critical	Phase 2	3	1m 11s	2026-09-26 20:18:19	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6108	61.9%	Low	Phase 1	1	10s	2026-09-26 20:18:12	Attack Stopped
10.0.0.18	SYN Flood	0.6322	94.0%	Critical	Phase 2	1	10s	2026-09-26 20:18:01	Attack persisted - escalating to time ban
10.0.0.10	UDP Flood	0.5576	92.0%	High	Phase 2	1	30s	2026-09-26 20:18:01	Ban Expired
10.0.0.5	Uncertain	0.6116	58.9%	Low	Phase 1	1	10s	2026-09-26 20:17:58	Attack Stopped
10.0.0.17	SYN Flood	0.6071	91.1%	High	Phase 2	2	41s	2026-09-26 20:17:54	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6283	94.1%	Critical	Phase 2	2	42s	2026-09-26 20:17:52	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6212	94.0%	Critical	Phase 2	1	41s	2026-09-26 20:17:51	Ban Expired
10.0.0.16	SYN Flood	0.6319	92.9%	Critical	Phase 2	2	41s	2026-09-26 20:17:49	Attack persisted - escalating to time ban
10.0.0.5	Uncertain	0.6133	58.9%	Low	Phase 1	1	11s	2026-09-26 20:17:48	Attack Stopped
10.0.0.5	Uncertain	0.6114	58.9%	Low	Phase 1	1	11s	2026-09-26 20:17:37	Attack Stopped
10.0.0.17	SYN Flood	0.6386	90.9%	Critical	Phase 2	1	11s	2026-09-26 20:17:24	Attack persisted - escalating to time ban
10.0.0.19	SYN Flood	0.6475	90.3%	Critical	Phase 2	1	11s	2026-09-26 20:17:21	Attack persisted - escalating to time ban
10.0.0.18	SYN Flood	0.6422	91.4%	Critical	Phase 2	1	10s	2026-09-26 20:17:20	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.16	SYN Flood	0.6378	92.9%	Critical	Phase 2	1	10s	2026-09-26 20:17:18	Attack persisted - escalating to time ban
10.0.0.8	Uncertain	0.6151	68.7%	Low	Phase 1	1	11s	2026-09-26 20:16:29	Attack Stopped

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE